
SYNTHESIZABILITY IS PER-MOLECULE; COST IS PER-LIBRARY: HUB-BATCHING FOR REACTION-GFLOWNETS

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ABSTRACT

Synthesizability-constrained molecular generation is evaluated one molecule at a time, but a laboratory makes a library. The cost of a library is not additive: compounds sharing an intermediate are cheaper together than apart, because the expensive part of making them has already been paid for. We therefore measure *distinct high-reward molecules delivered per fixed reaction budget*. Under this readout, a trained reaction-GFlowNet already contains the information needed to design a cost-efficient library: state flow at pre-terminal intermediates, recovered from a frozen checkpoint with no retraining or reward modification, reveals which cores are worth diversifying around. This procedure, which we call *hub-batching*, mirrors scaffold decoration in hit optimisation: make analogues around a high-value core so the cost of building it is amortized across many structurally distinct compounds. Hub-batching delivers a median $2.48\times$ more distinct high-reward molecules than reward-ranked selection across three reaction-grounded generators and four targets (23 of 24 comparable conditions; no case in which the baseline wins or ties), and, where only the selection rule differs on an *identical* candidate pool, $2.51\times$ against a cost-optimal solver. That solver is cheaper per candidate purchased, and one scoring oracle failed calibration against property-matched decoys. Reaction-grounded generation earns its keep at the library level even where per-molecule synthesizability does not distinguish it.

1 INTRODUCTION

Generative models for molecular design are scored, almost universally, one molecule at a time. A model is called synthesis-aware if its outputs are individually route-solvable or individually score well on a synthetic accessibility heuristic (Gao and Coley, 2020; Ertl and Schuffenhauer, 2009). That convention is a poor match for how compounds are actually made. A laboratory makes a library, not a single molecule, and the cost of a library is not additive in its members: if two target molecules share a late intermediate, the steps leading to it are paid for once and their cost amortized across everything downstream (Fromer and Coley, 2024). Per-molecule synthesizability is a property of a molecule; library cost is a property of a set; the two come apart.

We therefore measure *distinct high-reward molecules delivered per fixed reaction budget* — and a trained reaction-GFlowNet already contains what is needed to design a cheap library under it. Why the information is there at all is a property of the generator: a GFlowNet routes trajectories in proportion to reward, so the intermediates carrying the most flow are the ones whose descendants are both numerous and good — which is exactly what makes them worth building once and decorating many times. We find supporting evidence for this in Appendix B. The field’s central bottleneck is not model capacity but experimental data (Abolhasani and Kumacheva, 2023; Tom et al., 2024); the cost of synthesizing a designed library is the exchange rate between compute and wet-lab time, and it is not what current metrics measure.

Where the field stands. *Synthesis-aware generation* builds molecules by applying reaction templates to purchasable building blocks, so a route exists by construction rather than by inspection (Koziarski et al., 2024; Cretu et al., 2025; Seo et al., 2025; Gaiński et al., 2026; Gao et al., 2025). *Cost-aware selection* takes an existing candidate pool and chooses a batch, pricing shared intermediates explicitly (Fromer and Coley, 2024) and, in a recent extension, adding terms for diversity and for compatibility with parallel chemistry (Fromer et al., 2025). Between them sits an open dispute over whether the reaction Markov decision process (MDP) is needed at all: Kim et al. (2026) argue that template-based state and action spaces are unnecessarily rigid, and show that a sequence-based GFlowNet with soft synthesizability regularization achieves high synthesizable fractions with better flexibility and scale. Related work reaches similar conclusions by optimizing retrosynthesis models directly as an oracle (Guo and Schwaller, 2024; 2026). Measured per molecule, that argument holds. Our question is what happens one level up.

This work. We introduce *hub-batching*. Given any trained reaction-GFlowNet, we sample from a frozen checkpoint, recover the state flow $F(h)$ at pre-terminal intermediates by rearranging the detailed-balance condition (Bengio et al., 2023; Malkin et al., 2022), rank the pre-terminal intermediates behind the highest-reward candidates by that quantity, exhaustively enumerate every one-reaction child of the top-ranked ones, and walk the ranking greedily under a reward gate and a hard diversity constraint. There is no retraining, no reward modification, and no requirement beyond a reaction-grounded action space.

We note one thing that hub-batching is *not*. Parallel chemistry applies a single reaction across many reagents, and is the notion of batching that Fromer et al. (2025) optimize for. Hub-batching amortizes a shared *parent* across children reached by *different* reactions. Both save steps by reusing prior work, but they are different savings, and a library optimized for one is not automatically good under the other.

Our contributions are:

- **A library-level cost readout** with an explicit threat model, a comparability condition that makes fixed-budget comparisons well-posed, and an audit against an independent mixed-integer optimum (Section 2).
- **Hub-batching**, a post-hoc selection rule that reads the flow field of a frozen reaction-GFlowNet, evaluated across three reaction-grounded generators, a fragment-based GFlowNet as a cost-model control, and four targets (Sections 3-4).
- **A measured optimality gap and a set of severe tests**, two of which overturned our own earlier conclusions, including one of our four scoring oracles failing calibration (Section 5, Appendix D).

2 LIBRARY COST

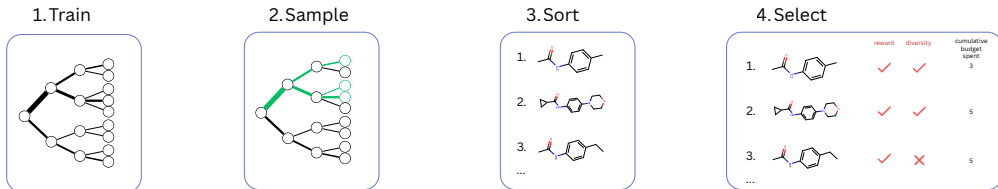
Definitions. Following previous literature for GFlowNet molecular design (Bengio et al., 2021; Zhang et al., 2023; Lau et al., 2024; Shen et al., 2025), we define a *mode* as a molecule that passes two filters: reward exceeds a per-target threshold, and Tanimoto similarity ($\tau = 0.5$, Morgan radius 3, 2048 bits) to every mode already accepted is below τ , grown greedily best-reward-first. We choose $\tau = 0.5$ to match the synthesis-aware work closest to ours (Shen et al., 2025) and stricter than the 0.7 in common use. Cost is charged *count-once*: each distinct reaction in the library’s shared synthesis DAG is paid for once, regardless of how many delivered molecules depend on its product. Our primary readout is the number of modes delivered at a fixed budget of **100 reactions**. We discuss filter choice and an external audit of the cost model in Appendix E.

Only runs that spent their budget are comparable. A fixed-budget comparison invites an obvious objection: a method that runs out of candidates before spending its budget will look cost efficient. We therefore label every run **budget-binding** (spent its budget with some candidates remaining) or **pool-exhausted** (ran out of qualifying candidates with budget unspent). Only budget-binding runs enter our analysis (Appendix E).

Threat model. The readout, number of modes at 100 reactions, is gameable from two directions: loosening what counts as a distinct molecule, or lowering what counts as a good one. Both are threshold choices, so rather than defend two particular values, we sweep both and report the surface (Figure 6). Our defaults ($\tau = 0.5$; reward at the 5% false-positive rate) are reported throughout for legibility, but hub-batching’s advantage ties or holds across $\tau \in [0.30, 0.90]$ and every reward cutoff we measured, with no crossover (Appendix E). Structural triviality is the one exploit a threshold sweep cannot rule out, so we additionally report synthetic-depth distributions (Appendix A). Taken to its limit, this exploit is the metric’s true degenerate optimum: purchasing diverse catalogue compounds costs zero reactions and yields many modes. This is distinct from concentrating on one intermediate with maximally diverse children, which is late-stage diversification and is what the method is for.

3 HUB-BATCHING

Best-Candidate (Previous)



Hub Batching

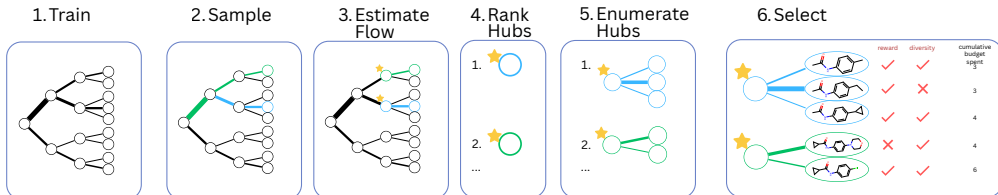


Figure 1: **Two ways to spend a reaction budget.** Reward-ranked selection (BEST-CANDIDATE, top) sorts the generator’s own samples and takes them greedily. Hub-batching (bottom) inserts one stage: it estimates state flow at pre-terminal intermediates, ranks them, and then *exhaustively enumerates* every one-reaction child of the top-ranked ones — so it *constructs* the pool it selects from rather than choosing among candidates it was given. **Both tracks end in the identical acceptance rule:** a molecule is taken only if it clears the reward gate and is Tanimoto $< \tau$ from everything already accepted. Only the *order* in which candidates are offered differs.

Procedure. Given a trained, frozen reaction-GFlowNet (Figure 1): (i) sample 30,000 trajectories; (ii) at each pre-terminal intermediate h with sampled terminal child x , recover the state flow; (iii) rank the pre-terminal intermediates of the highest-reward candidates by that estimated flow and keep the top 200 as *hubs*; (iv) exhaustively enumerate every one-reaction child of each selected hub and score it; (v) walk the hubs in flow order, accepting a child only if it clears the reward gate and is Tanimoto $< \tau$ from every already-accepted member; (vi) price the result with count-once, applied identically to every arm.

The flow identity. Step (ii) is the detailed-balance condition at edge $h \rightarrow x$, rearranged:

$$F(h) = F(x) \frac{P_B(h|x)}{P_F(x|h)}, \quad F(x) = \frac{R(x)}{P_F(\text{stop}|x)}, \quad (1)$$

computed in log space. We state this as a lemma rather than a theorem: it is a one-line rearrangement of a known condition (Bengio et al., 2023; Malkin et al., 2022). Its value

is interpretive. State flow is the reward mass of an intermediate’s terminal descendants, weighted by backward reachability—so ranking hubs by flow ranks them by the reward mass they capture, which is exactly the quantity a library designer wants to amortize an expensive parent against. The estimator we ship is the maximum over *sampled* children, the near-unbiased variant; alternatives averaging over enumerated children are biased upward, and we quantify this in Appendix B.

Baseline. BEST-CANDIDATE sorts the generator’s own samples by reward and accepts modes greedily. We credit BEST-CANDIDATE for any shared hubs or intermediates it happens to find, which corrected an earlier inflated result (Section 5). Best-candidate is a greedy upper bound on the savings a chemist would realistically capture.

4 RESULTS

Setup. Four generative models (RGFN, SCENT, RxnFlow, and a fragment-based GFlowNet) $\times$ four reward models spanning GPU docking (ClpP, CDK12–DDB1), a docking-score surrogate MPNN (sEH), and a bioactivity classifier (DRD2). Training details, building block library, and reward generator calibration are included in Appendix D.

4.1 MORE DISTINCT MOLECULES PER UNIT OF BENCH TIME

Hub-batching delivers a median $2.48\times$ more distinct high-reward molecules than best-candidate selection at a 100-reaction budget, over **23 of 24** comparable cells, range $1.70\text{--}3.64\times$, with **no cell in which the baseline wins or ties** (Figure 2); the advantage holds between 2.23 and $2.49\times$ at every budget from 50 to 200 reactions. We report 100 because it is the constraint a chemist actually has and the budget at which the competitor’s solver reliably converges (Appendix E), not because it is the most favorable point.

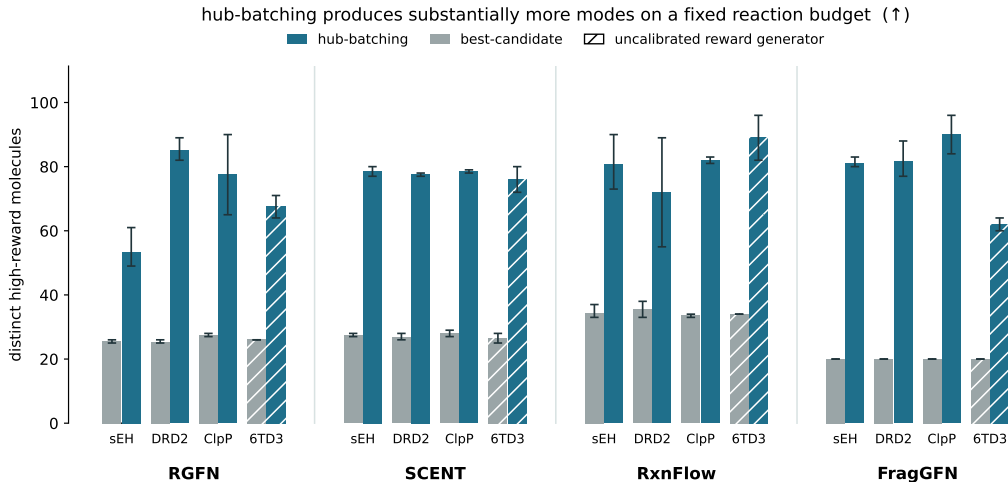


Figure 2: **Distinct high-reward molecules delivered at a fixed budget of 100 reactions**, hub-batching against reward-ranked selection, per generator–target cell; error bars show variance across random seeds. Pool-exhausted cells are hatched and excluded from ratios.

4.2 SELECTION, NOT ENUMERATION

A natural objection is that the generator, not the selection rule, is doing the work. We address this by holding the generator fixed (Gaiński et al., 2026) and comparing hub-batching against SPARROW, using SPARROW’s own diversity mechanism (Fromer et al., 2025) (Figure 5, Appendix C).

Measured by distinct modes, hub-batching delivers $3.26\times$ and $2.51\times$ more modes at a 100-reaction budget compared to SPARROW at $\lambda = 0$ and $\lambda = 1$, respectively, with $3.25\times$ and $2.5\times$ fewer reactions per mode. Measured by candidates, SPARROW with $\lambda = 1$ finds $1.21\times$ more candidates at $1.22\times$ fewer reactions per candidate.

Mechanism: a hard constraint beats a soft penalty when the budget binds.

SPARROW maximizes reward under a reaction cap with diversity entering as a penalty. When the cap binds, the cheapest available reward is another analogue hanging off an intermediate already paid for, and no affordable diversity weight outbids it: at $\lambda = 0$ the solver returns 99 compounds amounting to 24 and 26 distinct molecules on the two seeds. Hub-batching instead *refuses* any molecule within τ of one already accepted, and so pays to move elsewhere from the first selection onward.

Their knob is real: raising λ from 0 to 1 takes the solver from 24 to 33 distinct molecules on one seed and 26 to 32 on the other. Selection effort is itself a result: at a two-hour cap, 25 of 36 solver configurations did not converge, and neither a twelve-hour cap nor a relaxed tolerance fixed it, while our greedy walk answers the same problem on the same pool in about a second. We exclude non-converged rows, since each bounds the competitor from below rather than measuring it.

One result cuts against us. With a converged solve the competitor is *cheaper per candidate purchased* than we are (1.01 against 1.23) and buys slightly more of them; our claim is about cost per *distinct* molecule, where the ordering reverses (1.23 against 3.08).

Hub-batching also outperforms a full external pipeline (a route-less generator (Kim et al., 2026) planned by retrosynthesis and selected by SPARROW) winning on laboratory reactions while costing more candidates to score (Appendix F).

5 SEVERE TESTING AND LIMITATIONS

We tried to break these claims in several independent ways. Two attempts succeeded.

One oracle failed calibration. The CDK12-DDB1 oracle (PDB 6TD3) returns AUROC 0.688 against property-matched decoys (enrichment of $2.1\times$ where structurally matched decoys had suggested $82.9\times$). We exclude those cells from the headline and report the failure here; a correction will accompany our next round of training.

We corrected an inflation in our own favor. An earlier version of our cost model double-counted fragments in the baseline, reducing the naive-policy advantage from $\sim 1.81\times$ to $\sim 1.13\times$ after correction. All results use the corrected model, verified against SPARROW within 0.75%.

Limitations. Everything reported is *in silico*; ground truth is docking, not affinity. Yield and reaction-success probability sit outside the cost model (Appendix A). The solver receives roughly 13.1 alternative recipes per molecule on the competitor’s side against 1.0 on ours, so the optimizer has strictly more freedom against us than for us, which cuts against our reported advantage. Trained models also violate the detailed-balance identity at interior states (-6.05 to -14.03 nats at hubs); we analyze this in forthcoming work.

6 CONCLUSION

In the regimes we measure, reaction-grounded generation delivers more distinct high-reward molecules per unit of bench effort than route-less generation followed by post-hoc planning and cost-optimal selection over its highest-scoring pool—while costing more candidates to score. The selection signal was already inside the trained model; it had simply not been read. Future work should give cost-aware selectors pools that are diverse at construction time, so a selector is tested on its choices rather than on what it was handed. The field’s binding constraint is the cost of producing measured molecules. *Diverse* library synthesis cost is not downstream of the modeling problem. It is the problem.

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A YIELD OF MATERIAL

A reaction count treats every reaction as equal. Each has a *yield* — the fraction of material that survives the step — and compounding those along a route gives the share reaching the final compound. We ask whether hub-batching buys its advantage by choosing riskier chemistry.

It does not, and the mechanism is unambiguous (Figure 3). Per-template yields are taken from the SMALL library’s own reaction definitions (Gaiński et al., 2026), joined by exact SMARTS, and multiplied along each delivered molecule’s route. Mean *step* yield is **0.730–0.765** for **both arms on every cell**, so the two arms are drawing on the same chemistry. What differs is route *length*, and the direction follows it: on DRD2 hub-batching’s routes are shorter (2.66 against 3.37 steps) and more material survives, while on sEH they are longer (2.00 against 1.82) and less does. Yield is therefore a consequence of route length rather than of reaction choice, and it is a quantity a reaction count cannot see.

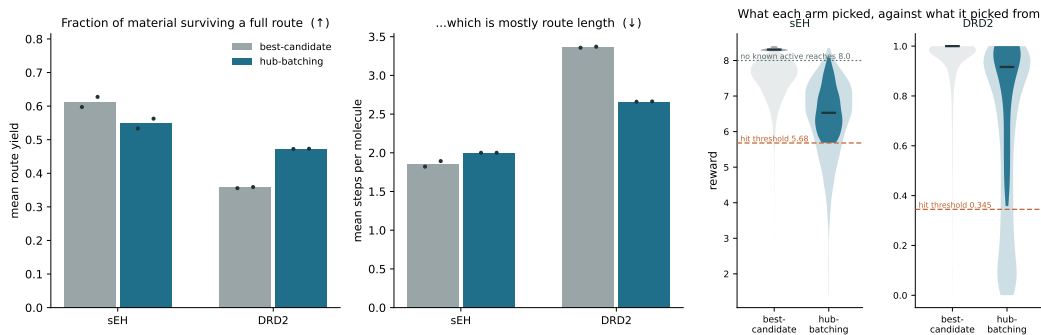


Figure 3: **How much material survives a route.** Mean route yield per cell, hub-batching against best-candidate, with the seeds shown individually. Mean *step* yield is **0.730–0.765** for both arms on every cell, so the differences are route *length*, not reaction choice. Read at the **5%-false-positive gates** (sEH 5.68, DRD2 0.345).

B FLOW FIELD CONTRIBUTES MOST ADVANTAGE

Floor. Figure 4 reports both halves. Reversing the ranking leaves the method unable to complete the library while burning the most compute; randomizing it completes but costs **1.53×** the reactions; the three arms that draw on high-flow intermediates land within **9%** of each other. The honest claim is therefore that **flow identifies the right neighborhood of intermediates, not the exact rank**, and we make no stronger one. One arm appears

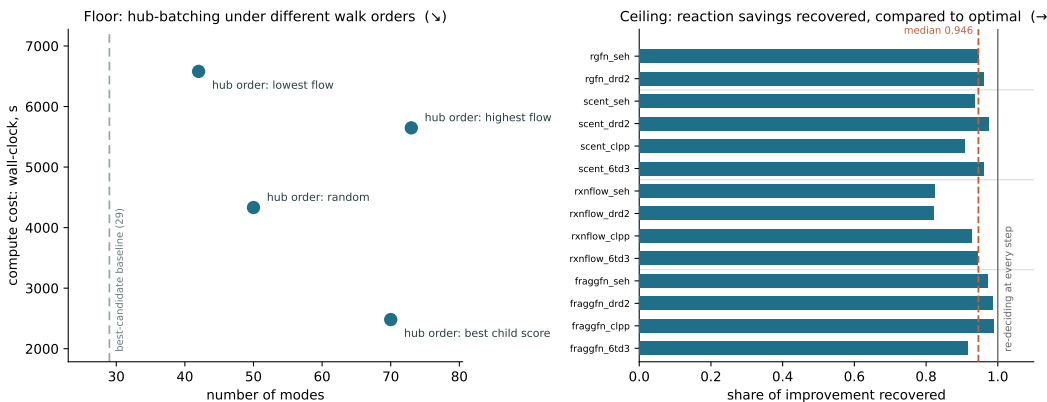


Figure 4: **Floor and ceiling for the hub ordering.** *Floor* holds everything fixed except the order hubs are visited in: SCENT with the sEH surrogate, 200 hubs, $R > 7.0$ and $\tau < 0.5$, one cell and one seed on a different checkpoint from Figure 2. *Ceiling* compares our one-shot flow ordering against an adaptive optimizer that re-scores every remaining hub at each step and takes the argmax marginal modes per marginal reaction. The reported figure is the share of the distance from the baseline to that optimizer which our ordering recovers.

to threaten this and on inspection supports it: ordering hubs by their single best molecule never reads flow, yet its selections sit at median flow rank 546 of 20,874 against random’s 11,043—because a GFlowNet routes trajectories through high-flow intermediates, so that ordering reads the flow field indirectly. Estimating flow is nearly free (0.2s for every observed intermediate; expanding one costs $\sim 3,000$ s), so the reward pre-filter in step (iii) is not a compute saving: letting flow select from all 20,874 observed intermediates instead changes the result by one mode (73 against 72).

Ceiling. To bound what a better ordering could buy, we built an adaptive optimizer that re-scores every remaining hub at each step by marginal modes per marginal reaction. Our one-shot ranking recovers a median 94.6% (range 82–99%) of the distance from baseline to that optimizer, costing a median 9.2% more reactions, or 10.9% fewer modes at $R = 100$. The optimizer buys the remainder at a price it does not advertise: $1.0\text{--}26.8\times$ more molecule scorings, since it cannot rank a hub it has not scored, and $0.6\text{--}79.8\times$ the selection wall clock. Part of the residual gap is not a saving at all—its libraries sit closer to the acceptance bar on 12 of 14 cells and are the most structurally collapsed of the three arms under a strict reward-blind clustering; forcing quality parity on the worst cell removes $\sim 37\%$ of its apparent advantage. It also does not rediscover our hubs (median flow rank 76 against a uniform null of 100; Jaccard 0.00–0.19), which is the same “neighborhood, not rank” conclusion from the opposite direction.

C SELECTION VERSUS ENUMERATION

D TRAINING DETAILS

We trained four models against four reward generators for 5,000 iterations with a reaction cap of 4 over a shared vocabulary of 418 building blocks and 112 reaction templates, following the SMALL library from Gaiński et al. (2026). The design is not balanced: sEH and DRD2 carry three seeds and ClpP and CDK12–DDB1 two, giving 40 generator–target–seed cells, of which 24 enter the headline. The fragment-based model is a **cost-model control, never a peer**: its “reactions” are fragment attachments rather than synthetic steps, and it is labeled as such wherever it appears.

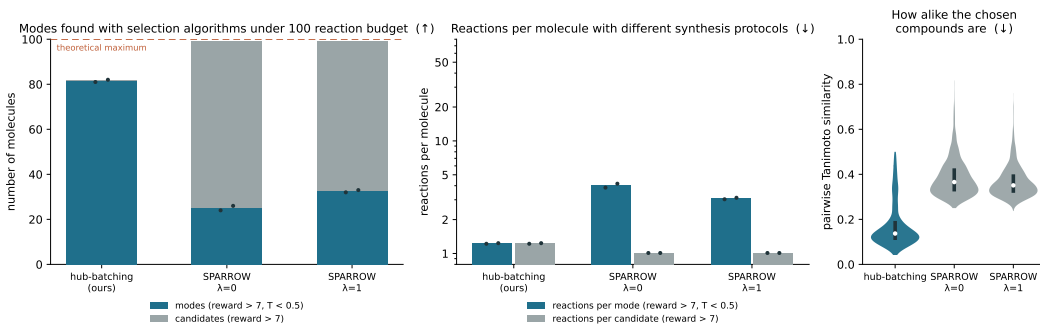


Figure 5: **Selection under a 100-reaction budget from the same candidate pool.** Both conditions select from the same set of candidates (41,817 from 64 hubs on seed 43; 39,969 from 73 on seed 44) so only the selection rule differs. We show SPARROW with its own diversity formulation ($\lambda = 1$) and without ($\lambda = 0$) (Fromer et al., 2025). Dots show results for individual random seeds. Discussed in Section 4.2.

Thresholds. Every reward gate is the 5% false-positive point on that oracle’s own actives-versus-decoys set: sEH 5.68, DRD2 0.345, ClpP −9.1. Figure 2 and Appendix A are read at those bars. The head-to-head comparisons (Figure 5, Table 1) are read at sEH 7.0, the bar at which prior sEH results are quoted, so they can be set against the published literature; the competitor’s pool is non-binding at both bars, holding 499 molecules above gate at each. CDK12–DDB1 keeps a provisional −2.0 and is excluded from every ratio, because its calibrated gate belongs to a different reward signal and so requires re-docking rather than re-gating.

Hub selection. Hub selection ranks the pre-terminal intermediates behind the highest-reward sampled candidates and expands the top 200 (64 in the head-to-head cell). Expanding 200 rather than 64 there moves our count by −2 and +1 modes across the two seeds and leaves the baseline unchanged at 27, since the baseline selects from the model’s own samples rather than from our enumeration.

The four targets are not interchangeable. ClpP is calibrated against property-matched decoys (AUROC 0.895). DRD2 is calibrated (0.949 held-out, 0.961 training-era, 18× enrichment) but *saturated*: 97.5% of generated molecules clear it, so DRD2 cells are effectively a diversity-only comparison. The sEH proxy is weaker and reported as such (0.76 against random molecules, 0.68 against property-matched decoys); we treat it as a scale, not an activity claim. The CDK12–DDB1 system *failed* calibration and its cells are excluded from the headline (Section 5). We report per-cell rather than pooled results so these differences stay visible.

E METRIC VALIDATION AND COST MODEL AUDIT

Both filters are load-bearing, and both cost us. Removing the reward gate makes hub-batching look 13% cheaper while 91% of the delivered library falls below the bar; removing the diversity constraint makes it look 12% cheaper while a 300-molecule library collapses to 39 distinct compounds. Priced per molecule worth keeping, those apparent savings cost 9.8× and 6.8× respectively, and 29× with both filters removed. The baseline is exactly indifferent to the reward gate, so the entire cost of that filter falls on the method that could otherwise exploit its absence.

The result is not an artifact of where the two thresholds sit. Every headline number is quoted at one setting of two knobs a reader can reasonably argue with: the reward bar, and the diversity cutoff. We therefore measure the whole grid rather than defend a point — 9 reward bars × 13 cutoffs, all four generators, at the paper’s own 100-reaction budget (Figure 6). Each cell is the ratio of the two arms’ reactions per distinct molecule, so

1.0 is parity; because the budget is fixed this is monotone in modes-at-100-reactions, the primary readout swept rather than a different metric. **No setting favors the baseline:** of 452 comparable cells, 0 fall below parity, and of the 400 drawn, 397 are strictly cheaper with 3 exact ties — all three at the strictest diversity settings. Median advantage is $2.57\text{--}3.06\times$ across the three reaction-grounded generators ($4.40\times$ for the fragment-based control, which is not a peer; Appendix D).

Two classes of cell are deliberately left uncoloured, because counting them would flatter us. *Pool-limited* cells (hatched) are ones where an arm could not spend the budget — either it ran short of library, or nothing cleared the bar at all — so its cost is measured over a shorter run and is not comparable. Both causes concentrate at the strict end, which is exactly where an unflagged surface would look most impressive.

The surface is close to flat in the reward bar (median $2.79\times$ at 4.0 against $2.56\times$ at 7.0) and narrows as the diversity requirement tightens ($2.86\times$ at $\tau = 0.7$, $2.49\times$ at our 0.5, $1.19\times$ at 0.3). That asymmetry is what the method predicts rather than a weakness of it: hub-batching earns its saving by amortizing one intermediate across many children, and children of a shared intermediate are similar by construction, so demanding extreme dissimilarity forces the walk off the hub and the amortization stops paying. Our operating point sits on the falling edge of that curve, not at its peak. This sweep varies thresholds on one cached enumeration per generator at seed 42; seed variance is Figure 5’s question. Each panel runs at its own generator’s operating configuration rather than a single shared one.

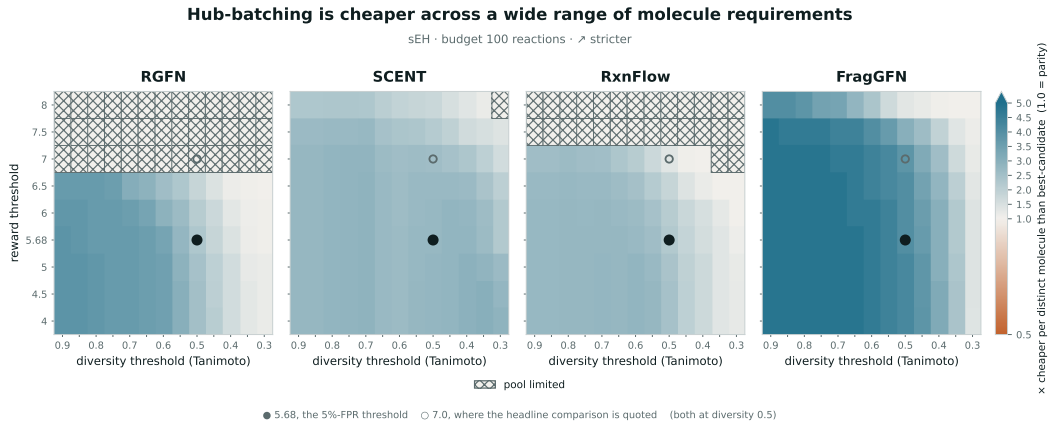


Figure 6: Hub-batching is cheaper across a wide range of molecule requirements. Sweeping both thresholds over their full range leaves the conclusion unchanged, so the result is not an artifact of where the thresholds were set. Reactions per distinct molecule, baseline $\div$ hub-batching, at a 100-reaction budget on sEH; 1.0 is parity. Rows are the reward bar, columns the diversity cutoff (Tanimoto), oriented so that up-and-right is *stricter* on both axes — the advantage narrows in that direction. Filled circle: 5.68, the 5%-FPR threshold. Hollow circle: 7.0, where the head-to-head comparison is quoted; both at cutoff 0.5. Hatched cells are pool-limited — an arm could not spend the budget — and are excluded, as are cells where no molecule clears the bar for either arm. Of 452 comparable settings, 0 favor the baseline. This varies thresholds, not seeds: one enumeration per generator, seed 42, so the magnitudes are not necessarily typical even though the ordering holds throughout. The fragment-based generator is a cost-model control, not a peer.

Default thresholds are strict To address the metric’s failure mode discussed in Section 2, we chose thresholds which were especially strict, as this gives the best case to *best-candidate*. Our choice of $\tau = 0.5$ is at the strict end of the 0.3–0.7 range used in the GFlowNet molecular literature (Lau et al., 2024; Shen et al., 2025). We calibrated the reward threshold with property-matched decoys to a 5% false-positive (FP) rate.

Table 1: **External comparison at a 100-reaction budget, sEH, seed 43.** Read at reward > 7.0 ; the competitor’s pool is threshold-non-binding, holding 499 molecules above gate at both bars we report.

Pipeline	Modes at $R = 100$ ($\uparrow$)	Ours $\div$ theirs
Kim et al. (2026) $\rightarrow$ retrosynthesis $\rightarrow$ SPARROW selection	46	1.76x
Kim et al. (2026) $\rightarrow$ retrosynthesis $\rightarrow$ greedy selection	55	1.47x
Gaiński et al. (2026) $\rightarrow$ Hub-batching (ours)	81	-

The cost model is audited externally. Against an independent mixed-integer optimum computed by SPARROW (Fromer and Coley, 2024) on the same molecules and routes, count-once places hub-batching 0.75% above provably optimal and the baseline 4.8–5.0%. Our selections sit closer to a global optimizer’s answer than the baseline’s do, so the advantage below is not an artifact of the accounting; where the two models disagree, ours is the conservative one.

Run status affects reported cost. Outside the budget-binding regime, a solver arm’s reported reaction count inflates relative to the cost of what it kept: in one pruned cell, 65 molecules priced at 247 reactions are reported against a budget usage of 300–387—so the gap between reported and kept cost is itself an exhaustion detector. Non-convergence is detected by wall-clock rather than by the solver’s status: CBC reports whichever incumbent it holds when the limit stops it as optimal, so a truncated run is indistinguishable from a converged one on status alone.

F AGAINST A FULL EXTERNAL PIPELINE

To confirm this is not an internal comparison, we run a full comparison of hub-batching against an external candidate generation pipeline. We plan its molecules with a retrosynthesis model, and select molecules over the resulting pool by using either SPARROW or a diversity-aware greedy rule (Table 1).

We report the asymmetries rather than equalizing them. The competitor’s selector chose from 438 routed candidates, drawn from its own frozen policy at a cost of 41,346 eval-phase scoring calls, and 2.25 h of retrosynthesis it must pay and we do not; our selector chose from 41,817 enumerated children above the gate, at 167,833 scoring calls and no route planning. Those draws cost 10,271 / 41,346 / 10,342 calls for seeds 42/43/44 against a 10,048-call training budget, so we report them per seed rather than averaging: the seeds differ fourfold in how much sampling the pool required. **Hub-batching outperforms on laboratory reactions and loses on the number of molecules you must score**, which is the structural consequence of constructing a candidate pool rather than selecting from a given one.

A caveat on the competitor’s pool. Its candidates are the highest-scoring molecules its generator produced. That is the pipeline as normally run, but it concentrates the pool on similar chemistry before the solver sees it, and pool construction rather than the selector may be what bounds the result: a diversity-enforced pool changes the picture materially. We report the score-ranked construction because it is the default, and treat the comparison as unsettled rather than closed.